1. p = Probability that the bomb strikes the target $= 50\% = \frac{1}{2}$

Let n be the no. of bombs which should be dropped to ensure 99% chance or better of completely destroying the target.

$\Rightarrow$ probability that out of n bombs at least two strikes the target is greater than 0.99.

Let X be a r.v representing the no. of bombs striking the target. Then $X \sim B(n, p = \frac{1}{2})$

$$p(x) = P(X = x) = {}^nC_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x}$$

$$= {}^nC_x \cdot \left(\frac{1}{2}\right)^n, \quad x = 0, 1, 2, \ldots n.$$

$$P(X \geq 2) \geq 0.99 \Rightarrow 1 - P(X \leq 1) \geq 0.99$$

$$\Rightarrow 1 - (p(0) + p(1)) \geq 0.99$$

$$\Rightarrow 1 - ({}^nC_0 + {}^nC_1)\left(\frac{1}{2}\right)^n \geq 0.99$$

$$\Rightarrow 0.01 \geq \frac{1+n}{2^n} \Rightarrow 2^n \times 0.01 \geq 1 + n$$

$$\Rightarrow 2^n \geq 100 + 100n$$

By trial method: $\boxed{n = 11}$ (A)

$\therefore$ The min no. of bombs needed to destroy the target completely $= 11$ Ans.

2. $\lambda = 1$, $P(X = x) = \dfrac{e^{-\lambda}\lambda^x}{\lfloor x} = \dfrac{e^{-1}}{\lfloor x}$, $x = 0, 1, 2, \ldots$

$$E(|X-1|) = \sum_{x=0}^{\infty} |x-1| p(x) = e^{-1} \sum_{x=0}^{\infty} \frac{|x-1|}{\lfloor x}$$

$$= e^{-1}\left[1 + \frac{1}{\lfloor 2} + \frac{2}{\lfloor 3} + \cdots\right]$$

$$\frac{n}{\underline{|n+1}} = \frac{n+1-1}{\underline{|n+1}} = \frac{1}{\underline{|n}} - \frac{1}{\underline{|n+1}}$$

$$E(|X-1|) = e^{-1}\left[1 + \left(1 - \cancel{\frac{1}{\underline{|2}}}\right) + \left(\cancel{\frac{1}{\underline{|2}}} - \cancel{\frac{1}{\underline{|3}}}\right) + \left(\cancel{\frac{1}{\underline{|3}}} - \frac{1}{\underline{|4}}\right) + \cdots\right]$$

$$= e^{-1} \cdot 2\times 1 = \frac{2}{e} \times \text{standard deviation.}$$

[C]

(3) $$P(X=10) = {}^{9}C_2\, p^2 q^7 \cdot p = {}^{9}C_2\, p^3 q^7$$

$$= \frac{9\times 8}{2\times 1} \times (0.4)^3 \cdot (0.6)^7$$

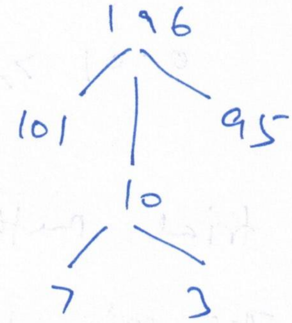
$$= 0.0645 \quad \text{Ans}$$

$$P(X=x) = {}^{x-1}C_{r-1}\, p^r q^{x-r}, \quad x = r, r+1, r+2, \ldots$$

$x=10,\ r=3,\ p=0.4,\ q=0.6$

(A)

(4) let X: no of female candidates



$$P(X=7) = \frac{{}^{101}C_7 \times {}^{95}C_3}{{}^{196}C_{10}}$$

$$= 0.130$$

(A)

(5) $$P(X=3) = pq^{x-1} = pq^{3-1}$$

$| x=3$

$$= pq^2$$

$| p=\frac{2}{3},\ q=\frac{1}{3}$

$$= \frac{2}{3}\cdot\left(\frac{1}{3}\right)^2$$

$$= \frac{2}{27} \quad \text{Ans.}$$

$$= 0.074$$

(A)

6. $T \sim$ Exponential distribution with mean $\frac{1}{2}$ $\therefore E(X) = \frac{1}{2} = \frac{1}{\lambda}$

$\Rightarrow \lambda = 2$

$\therefore f(x) = 2e^{-2x}, \quad x > 0$

$= 0, \quad$ otherwise.

$$P(1 < T < 2) = \int_1^2 2e^{-2x}\,dx$$

$$= \left[-e^{-2x}\right]_1^2$$

$$= \boxed{e^{-2} - e^{-4}}$$

7. $X \sim N(165, 9)$

$\mu = 165$, $\sigma^2 = 9$ $\therefore \sigma = 3$

$P(X < 162)$

$$= P\left(\frac{X - 165}{3} < \frac{162 - 165}{3}\right)$$

$= P(Z < -1)$, where $Z \sim N(0,1)$

$= P(Z \leq -1)$

$= \boxed{\Phi(-1)}$

8. The total time, T, the batteries will last is a sum of two exponential i.i.d. random variables. Therefore, T follows a Gamma distribution with parameters $n = 2$ and $\lambda = \frac{1}{2}$.

$$\therefore P(T > 4) = \int_4^\infty \lambda^2 t e^{-\lambda t}\,dt = \boxed{3e^{-2}}$$

9. $X \sim Exp(1)$ $\therefore f(x) = e^{-x}, \quad x > 0$

Due to the memoryless property of exponential distribution,

$$P(X > 10 \mid X > 5) = P(X > 5)$$

$$= \int_5^\infty e^{-x}\,dx$$

$$= \left[-e^{-x}\right]_5^\infty$$

$$= \boxed{e^{-5}}$$

10. $X \sim$ Uniform distribution in $[20, 40]$

$$\therefore E(X) = \frac{a+b}{2} = \frac{20+40}{2} = \boxed{30}$$

$$\sigma^2 = V(X) = \frac{(b-a)^2}{12} = \frac{(40-20)^2}{12} = \frac{(20)^2}{12} \quad \therefore \boxed{\sigma = \frac{20}{\sqrt{12}}}$$